

DECENTRALIZED DIGITAL CURRENCY PROTOCOL

DDCP Architecture — Technical Overview

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DDCP Foundation, Inc.
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1. What Is DDCP

DDCP is Better Money: a protocol rooted in unconditionality, privacy, protection from arbitrary seizure, and value preservation. DDCP's deployment as a narrow, open-source protocol on Solana supports the issuance of stablecoins that combine privacy, unfreezability, seizure resistance, and trustworthiness — a combination not available from any other stablecoin currently in existence.

The protocol is narrow by design: it executes value transfer with privacy-preserving properties. No general-purpose, user-deployable smart contract execution is supported. This boundary is deliberate: any user-deployable contract on the value-transfer layer would create a vector through which conditions, freeze capability, or seizure capability could be reintroduced.

Unfreezability and seizure resistance

No administrative key exists, at any layer of the protocol, that can freeze, restrict, or move the contents of a self-custody wallet. This is an architectural fact rather than a policy commitment: the DDUSD mint initialization specification includes no Freeze Authority and no Freeze Account extension — the capability to freeze a token account is absent by design, and neither a revoked Freeze Authority field nor freeze-related extensions can be added or re-enabled on an existing Token-2022 mint post-initialization (§3, C1). AML/CFT compliance is handled entirely at the Financial Intermediary boundary (§5): the protocol provides the infrastructure, and the obligation sits with the regulated entities crossing the fiat boundary.

Privacy

Account balances and the amounts of individual transfers are concealed from on-chain observers by default for self-custody wallets, using the Confidential Balances suite of Solana Token Extensions (via Token-2022

program) and ZK ElGamal proofs verified natively by Solana (§3, C2). What a holder owns and what they transact is not public information.

Unconditionality

DDCP tokens, once held, carry no embedded restrictions on how they can be used, by whom, or for what purpose. The protocol implements no allowlists, denylists, or other form of programmable conditionality on transfers. A token in a self-custody wallet behaves identically for every holder, everywhere — the property physical cash has, and that most digital payment systems do not.

Value preservation

Every token issued under DDCP is backed by reserve assets, under a structure that is independently verifiable rather than dependent on trust in any single party (§3, C4). Reserve management is performed by a foundation structurally separate from the issuer, with independently audited reserve holdings and attestation results published on-chain at defined intervals — trustworthiness as a property anyone can verify, not a claim resting on any institution's reputation.

The protocol has no built-in yield distribution mechanism to token holders. Whether and how an Operator passes yield to holders is a deployment-level commercial and regulatory determination, made independently for each token instance.

2. DDUSD — Product and Competitive Position

DDUSD, the first token deployed on DDCP, is a USD-pegged stablecoin: each token is issued by DDUSD, LLC and backed 1:1 by reserve assets held independently under the RMF (§3, C4). DDUSD inherits DDCP's core properties (§1) for a USD-denominated instance: architectural unfreezability, balance and transfer-amount privacy, unconditional use, and full, independently-attested backing.

DDUSD, LLC is pursuing a Permitted Payment Stablecoin Issuer (PPSI) license from the OCC under the GENIUS Act. As a Permitted Payment Stablecoin, DDUSD will make available to US persons a set of characteristics not offered by any other stablecoin currently available to them. Reflecting current GENIUS Act constraints, DDUSD does not pass reserve yield through to holders — a deployment-specific instance of the

general mechanism described in §1. The comparisons that follow describe currently available alternatives as presently configured.

USDC and USDT, the dominant USD stablecoins on Solana, both maintain administrative keys that can freeze any token account including self-custody wallets, and both publish every balance and transfer amount on-chain. PYUSD has the Confidential Balances extension provisioned on its mint but inactive — amounts and balances remain public; activation requires a deliberate opt-in step that PayPal has not surfaced in its user interface. PYUSD also has a Permanent Delegate, a Paxos-controlled address able to transfer or burn tokens from any account without the holder's consent — a unilateral asset-movement capability, not merely a freeze — and a Transfer Hook allowing Paxos to run custom logic on every transfer. Its Confidential Transfer configuration includes a live Paxos auditor decryption key that, if activated for an account, would let Paxos decrypt that account's balances and amounts. DDUSD has confidential balances on by default, no Permanent Delegate, and no auditor decryption key.

Aleo's stablecoin privacy architecture wraps existing stablecoins, such as USDC, in a private representation at a protocol level. The underlying issuer's freeze key remains intact on the source chain — a holder of Aleo-wrapped USDC in a private self-custody wallet on Aleo is still exposed to Circle's freeze capability on Ethereum. DDUSD is issued natively on Solana: there is no upstream exposed token and no upstream freeze vector. Arcium is a decentralized MPC network integrated with Solana that enables computation over encrypted data — but its privacy guarantee depends entirely on the correct and continuous operation of Arcium's own off-chain node network: if that network is unavailable, compromised, or behaves incorrectly, the privacy property fails regardless of Solana's state. DDUSD's Confidential Balances require no such external dependency: the ZK ElGamal proof is generated client-side and verified by Solana's native proof program, making the privacy guarantee self-contained within the same trust boundary as the token itself.

3. Protocol Architecture

The architecture has four components.

C1 — Issuance

DDUSD is implemented as an SPL token on Solana's Token-2022 program. Minting and burning authority sit in a separate issuer operations program, isolated from the transfer path.

A mint requires co-signatures from both the Issuer (DDUSD, LLC) and the RMF (introduced in C4); the co-signature program checks the current on-chain attestation record and will not countersign a mint that would push circulating supply above attested reserves. Symmetrically, the RMF does not release reserve assets until the corresponding on-chain burn is confirmed.

The issuer operations program includes a mint-only kill switch: minting can be paused but burns and user-to-user transfers remain live regardless of pause state, so a pause cannot be used to trap funds.

This is a selective-immutability design, expressed through the DDUSD mint's authority settings and extension layout. The mint's Freeze Authority is permanently set to None, and the Token-2022 Freeze Account extension is entirely absent from the mint initialization layout; consequently, no key — not DDUSD, LLC's, not DDCP's, not anyone's — can ever freeze a token account through the standard Token-2022 freeze mechanism. The mint's Mint Authority remains active, held by the co-signature program described above, since issuance must continue. DDUSD, LLC's operations program maintains its own managed upgrade authority for operational changes — such as onboarding Financial Intermediary counterparties — but that authority has no reach into the mint's authority setting or the Token-2022 transfer path. The result is an architectural impossibility for any administrative key to freeze self-custody token accounts, rather than a policy commitment: the component that could carry such a capability was never built into the immutable mint structure.

C2 — Privacy

DDUSD uses the Token-2022 Confidential Balances Token Extension. Transfer amounts and account balances are hidden from on-chain observers using ZK ElGamal proofs, verified natively by Solana.

Two balance states exist. Self-custody wallets hold a confidential balance by default — transfer amounts and balances between self-custody wallets are hidden on-chain. DeFi interactions (Orca, Jupiter, Kamino, MarginFi, and equivalent protocols) use a public balance; a user who wants to participate in DeFi moves from confidential to public balance through a

deliberate unshielding step before interacting with an AMM, DEX, or lending protocol (§4). This preserves full composability with Solana's DeFi ecosystem without fragmenting liquidity between a confidential pool and a public pool.

DDUSD's Confidential Transfer mint configuration does not provision an issuer-held auditor decryption key. Unlike some competing implementations (§2), no key exists that could be activated to decrypt user balances. Compliance runs through the Financial Intermediary boundary (§5), not protocol-level issuer decryption. Sender and recipient addresses remain visible on-chain.

C3 — Consensus

DDCP, and all its token instances, run on Solana's existing, internationally distributed validator network — a large, already-decentralized base with sub-second finality and throughput well above what DDUSD requires.

How users cover transaction fees is under evaluation. Solana's standard fee model requires a SOL balance to execute transactions; a fee-payer arrangement — where a separate account covers the SOL cost on a user's behalf — is one application-layer option. A second option under evaluation is the Transfer Fee Extension: a per-transfer DDUSD fee defined as a percentage of the transfer amount, capped at a maximum fee — combined with an application-layer restriction that prevents users from injecting volatile priority fees — so it approximates Solana's roughly flat per-transaction cost for ordinary transfers without taking a disproportionate share of very small ones.

C4 — Reserve Custody

The DDCP Reserve Management Foundation (RMF) centralizes the reserve management function — custody relationships, rebalancing execution, attestation, and governance oversight — while each deployment's reserve assets remain independently ring-fenced and attested. Structured as a foundation separate from the Issuer, with a multi-signature authorization structure for material reserve actions, the RMF's mandate is value preservation at par. Its scope is limited to reserve management and yield distribution; it has no authority over the protocol's technical parameters.

Reserve assets are held under an orphaned bankruptcy-remote special purpose vehicle that legally isolates them from any parent entity. The majority are held in true custody — not deployable by the custodian as loanable funds, with the custodian earning a fee rather than a spread. A portion is held as bank deposits to meet short-term redemption liquidity needs. Overall reserve composition is governed by the RMF against publicly specified criteria for duration, liquidity, and issuer credit quality; reserve adequacy is independently audited, with attestation results published on-chain at defined intervals.

The mint/burn co-signature controls described in C1 are the primary mechanism through which reserve adequacy is enforced at the transaction level. The co-signature program checks the current on-chain attestation record before countersigning any mint — a mint that would push circulating supply above the last attested figure cannot execute. Between scheduled attestation events, when the on-chain record reflects a prior snapshot rather than the current reserve position, the RMF co-signature serves as the operative real-time control: the RMF will not countersign a mint it cannot itself verify is supported by current holdings. Reserve assets are not released until the corresponding on-chain burn is confirmed.

The RMF's full governance specification — legal structure, multi-signature composition and decision rules, attestation cadence, rebalancing triggers, and conflict-of-interest framework — is to be documented separately.

4. Wallets and DeFi Access

The no-freeze and confidential-by-default properties (§3, C1 and C2) are properties of the token program configuration, not of any particular wallet — they hold for any standard Solana wallet that supports Token-2022.

Native UI support for Confidential Balances varies across the ecosystem. While software wallets like Phantom and Solflare provide front-end flows for managing encrypted states, hardware wallets and advanced implementations may require the Solana CLI tool suite as a fallback to generate the necessary client-side zero-knowledge proofs. On- and off-ramp access is provided by Financial Intermediary partners (§5).

As a narrow protocol, DDCP provides settlement and privacy properties for the token, but it does not host on-chain credit infrastructure; credit ecosystems build on top of that. DDUSD holders reach Solana's DeFi

ecosystem directly (liquidity, lending, routing via Orca/Raydium/Meteora/Kamino/MarginFi/Jupiter, with the unshielding step from §3, C2).

5. The Financial Intermediary Boundary

A Financial Intermediary (FI) is any entity that crosses the fiat boundary — converting between fiat currency and DDCP tokens in either direction — or holds custody of the private keys authorizing transactions from a user's account. The test is functional: what matters is whether either condition is met, not how an entity describes itself.

Actor	FI?	Basis
Fiat on/off-ramp operator	Yes	Crosses fiat boundary; primary AML/CFT obligation bearer
Custodial wallet provider	Yes	Holds key custody
Non-custodial wallet (keys on user's device)	No	User signs unilaterally; wallet holds no key material and crosses no fiat boundary
MPC wallet — server as co-signer	Yes	Server is a required co-signer.
MPC wallet — server as recovery only	No	User signs unilaterally in normal operation. Server as a recovery mechanism only.
Issuer (DDUSD, LLC)	Yes	Crosses fiat boundary at the wholesale issuance level
DDCP Foundation, Inc.	No	Stewards the open-source protocol specification
DDCP Reserve Management Foundation (RMF)	No	Manages reserve assets only — no user wallet interaction, no fiat-boundary crossing, no key custody

AML/CFT compliance sits with FIs, not with the protocol. The co-signature mint/burn architecture (§3, C1) gives the Issuer a compliance capability that operates through FI coordination: working with an FI acting under a lawful order, the Issuer can decline to mint new tokens to a person the FI has

identified as the subject of that order, decline to process redemption for tokens presented by or on behalf of that person, and execute an on-chain burn of tokens held in a custodial relationship at that FI when the FI presents them under the order, reducing outstanding issuance with an on-chain record. These capabilities apply to tokens held in custodial FI relationships and operate through coordination with the FI holding the account, not through any unilateral protocol-level action on a self-custody wallet.

The two-balance-state design (§3, C2) has a direct AML/CFT consequence: DeFi-pool transactions occur in the public balance, so on-chain amount data is available to any observer, while self-custody-to-self-custody transactions occur in the confidential balance, so on-chain amounts are hidden. Either way, the FI boundary carries the compliance obligation — on- and off-ramp operators observe and report fiat-side transaction data, which is fully visible regardless of the on-chain confidentiality state.

6. Realizing the Vision

From its first day, DDCP's deployment as a narrow, open-source protocol on Solana realizes several of the central commitments declared in the Manifesto. No administrative key exists, at any layer, that can freeze, restrict, or move funds out of a self-custody wallet (§3, C1) — unfreezability as an architectural fact. Account balances and individual transfer amounts are concealed from on-chain observers for self-custody wallets (§3, C2) — privacy as an operating default, not an opt-in feature. Consensus and governance run on Solana's existing, internationally distributed validator network (§3, C3) — a large, already-decentralized base rather than a new network that would need to earn that distribution over time. And every token instance is backed 1:1, with reserve custody and attestation structurally separated from the issuer (§3, C4) — value preservation at the level of the stablecoin, reserve attestation results published on-chain.